Tumors Systematically Bias Brain Age Prediction: Towards Lesion-Agnostic Estimation via Inference-Time Inpainting

Kayra Özdemir
Department of Radiotherapy, Amsterdam UMC, location VUmc
BSc. Artificial Intelligence, Faculty of Science, VU Amsterdam

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Background

Brain age models are **trained on healthy MRI with chronological age as supervision signal**, thereby learning to predict age solely from brain structure. The deviance between their prediction and the actual age of the patient is used as a means of a biomarker. [1]

$$\text{Brain Age Gap (BAG)} = \textit{Predicted Age} - \textit{Chronological Age}$$

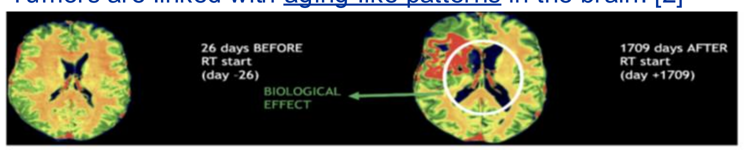
$+$ BAG $\longrightarrow$ brain appears older

$-$ BAG $\longrightarrow$ brain appears younger

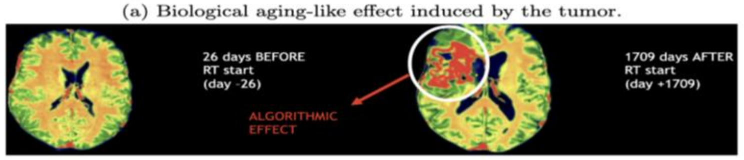
Brain aging is generally characterized by **tissue loss.**

Brain tumors are lesions that alter MRI appearance through:
- **Focal artifacts** (i.e., bounds of the tumor)
- **Diffuse anatomical remodeling** (e.g., global atrophy)

Tumors are linked with aging-like patterns in the brain! [2]



(a) Biological aging-like effect induced by the tumor.



(b) Algorithmic bias effect arising from presence of lesion voxels.

Problem

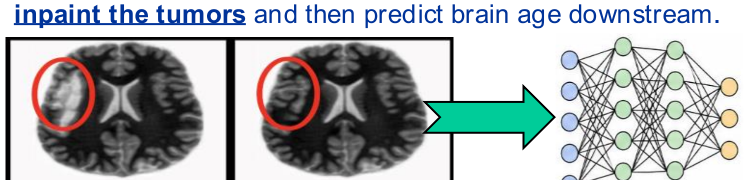
1. As the training distribution contains *only* healthy MRI, models may predict **inaccurately** for brains with lesions (e.g., tumors).

2. These models are **required** to be trained on healthy distributions so that supervised learning does not suffer from *data contamination*: the model needs to learn what a healthy brain looks like for the deviation to be used as a biomarker.

3. We **can not directly quantify the perturbation effect** of tumors **on real patients**, as the algorithmic error value is unknown, real tumors would have induced extra aging, and there is no counterfactual healthy prediction for comparison.

Goal & Core Idea

The goal is to devise a **tumor-agnostic** brain age prediction method without relying on training or augmentation techniques (inference-time strategy). The core idea is to **inpaint the tumors** and then predict brain age downstream.

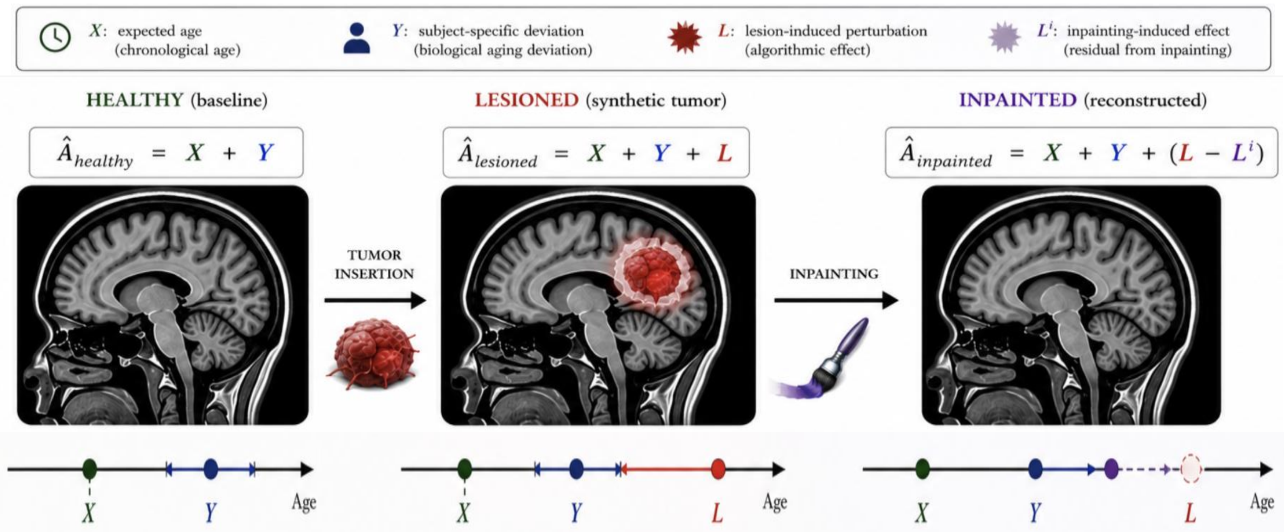


Research Questions

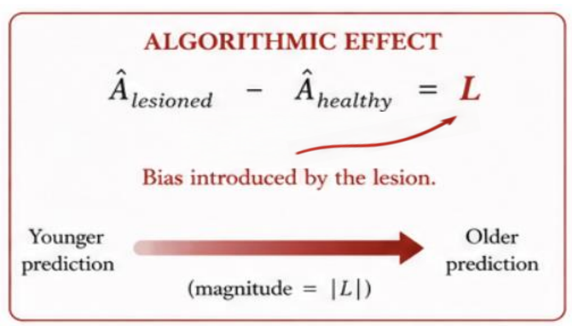
The thesis is split up into two phases: first, proving that tumors bias brain age models systematically; second, evaluating whether inpainting is a solution, through two RQs:

- **RQ1**: Do tumors induce systematic bias on predictions produced by brain age models?
- **RQ2:** Does inpainting recover biased predictions toward the ground-truth prediction that would have been made for the healthy version of that brain?

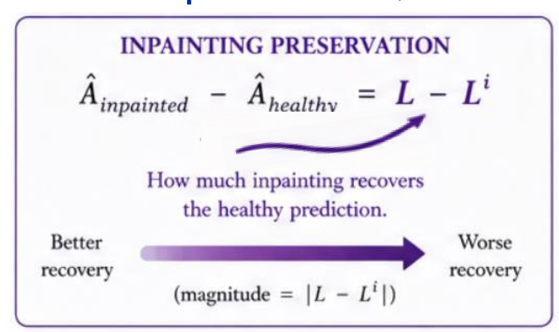
Experimental Framework



Experiment 0 / RQ1



Experiment 1 / RQ2



This setup:
- Addresses the issue of not being able to quantify perturbation effect (L) on real patients, by enabling **controlling for the tumor**. This also eliminates extra aging caused by real tumors.
- Evaluates whether inpainting is a solution by using the generated synthetic tumored output from the first experiment (RQ1) with an **already available healthy ground-truth prediction**. Showing that inpainting solves the problem for synthetic tumors would mean the method is directly applicable to patients with real tumors for robust and reliable brain age prediction.

Methodology

Brain Age Models: BrainAgeNeXt [3], Joos Multi-Task [4]

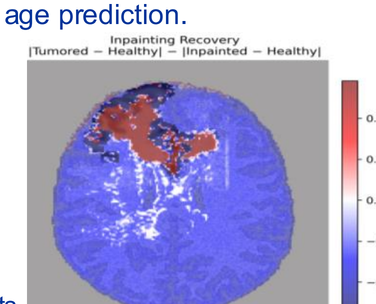
Synthetic Tumor Generators: CarveMix [5], GliGAN, [6], USB [7]

Inpainting Modules: Brain-ID [8], FastSurfer-LIT [9], USB [7]

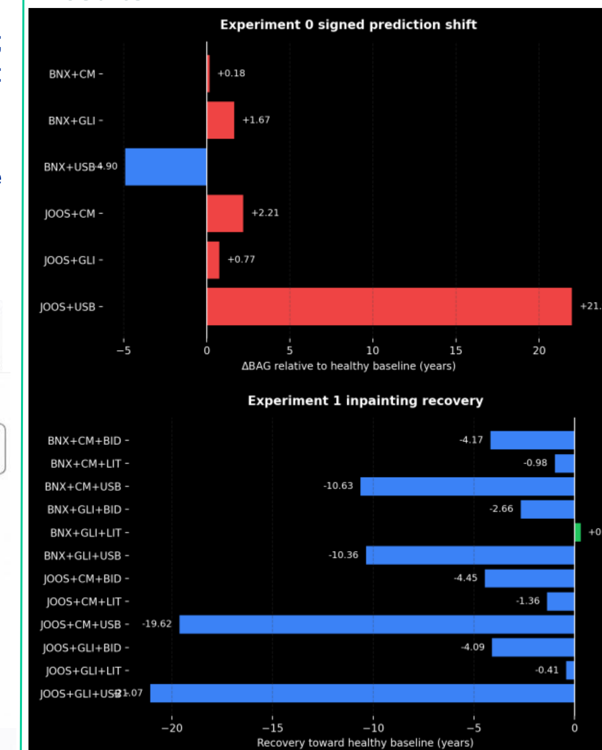
Data: IXI [10] for healthy T1 MRI, BraTS for tumor masks [11] (n=547)

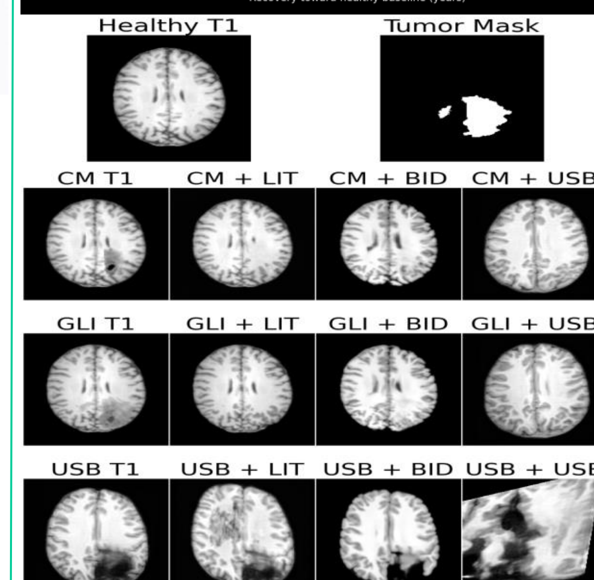
Evaluation: ΔBAG, |ΔBAG|, inpainting recovery

Statistical Analysis: Paired Wilcoxon signed-rank tests, permutation tests



Results





Interpretations

1. Brain tumors systematically bias brain age predictions, demonstrating that brain age models are not lesion-agnostic. This raises concerns about previous works focused on brain age prediction in pathological populations.

2. Current inpainting methods do not reliably recover baseline healthy anatomy, and successful lesion removal does not guarantee preservation of underlying tissue.

References

[1] Cole et al., 2017, [2] Claes et al., 2007, [3] La Rosa et al., 2024, [4] Joos et al., 2026, [5] Zhang et al., 2023, [6] Jain et al., 2025, [7] Wang & Liu, 2025, [8] Liu et al., 2024, [9] Pollak et al., 2025, [10] IXI Consortium, 2007, [11] BraTS Challenge, 2024